MAC TUTORIAL — BEGINNER'S GUIDE

PHP Environment on macOS

Step-by-step installation and usage guide

macOS 14+ — Mac Intel & Apple Silicon (M1/M2/M3/M4/M5)

🌐 Apache 2.4 | ⚙️ PHP 7.4 | 🗃️ MariaDB 10.6 | 🔧 phpMyAdmin | 🐳 Portainer

🌐 PHP Site	`http://localhost:8080`
🗄️ phpMyAdmin	`http://localhost:8081`
🐳 Portainer (GUI)	`http://localhost:9000`
🏠 Dashboard	`http://localhost:8082`

Table of Contents — Mac

1 Install Docker Desktop on Mac p. 3

- Mac Intel or Apple Silicon — how to tell?
- Download and installation
- Verify that Docker is ready

2 First-time setup & launch p. 4

- Run install.sh (one-time macOS setup)
- Launch the environment with the launcher

3 Access the web interfaces p. 5

- Dashboard (8082), PHP Site (8080), phpMyAdmin (8081)
- Portainer — create your account (important!)

4 Working with PHP p. 7

- The htdocs folder — your workspace
- Create and test a PHP file
- Connect to the database

5 Manage the database with phpMyAdmin p. 8

6 Stop and restart services p. 9

7 Mac Troubleshooting p. 10

- Portainer timeout — solution
- Common issues
- Credentials summary

About this tutorial

This guide is designed for beginners. All operations are performed through graphical interfaces. No complex command line is required after the initial setup.

CHAPTER 1

1 Install Docker Desktop

To be done once on your macOS

🍎 What type of Mac do you have?

Before downloading, identify your processor. The two Mac types require different versions of Docker.

🔍 **How to tell?**

Menu **Apple ()** → **"About This Mac"**

- You see **"Intel Processor"** → Mac Intel
- You see **"Apple M1/M2/M3/M4/M5 Chip"** → Mac Apple Silicon

⬇️ Download and installation

🖥️ Mac Intel

1. Go to **docker.com**
2. Click **"Download for Mac"**
3. Choose **"Intel Chip"**
4. Open the `.dmg` file
5. Drag Docker → Applications
6. Open Docker Desktop

⚡ Mac Apple Silicon (M1/M2/M3/M4/M5)

1. Go to **docker.com**
2. Click **"Download for Mac"**
3. Choose **"Apple Chip"**
4. Open the `.dmg` file
5. Drag Docker → Applications
6. Open Docker Desktop

✅ Verify that Docker is ready

Docker Desktop places a **whale icon** 🐳 in the menu bar (top right of the screen).

Color meanings

Green = Ready	Docker is running — you can work
Yellow = Starting	Wait a few seconds
Red = Error	Restart Docker Desktop

⚠️ **Docker must be green before continuing**

Docker Desktop must remain open during your entire work session. If you close it, all services will stop.

CHAPTER 2

2 First-time setup & launch

One command to set everything up, then double-click to launch

🔧 One-time setup — `install.sh`

macOS blocks downloaded files for security (Gatekeeper). This script removes those blocks, sets permissions, and checks Docker. **You only need to do this once.**

1. **Open Terminal**
   Launchpad → search "Terminal" → click on it

2. **Paste this command and press Enter:**

```
cd ~/Desktop/PHPenv && bash install.sh
```

3. **Wait for "Setup complete!"**
   The script removes security blocks, sets permissions, and verifies Docker is installed

✅ **After this step**
You can double-click any `.command` file without security warnings. You never need to run `install.sh` again.

🚀 Launch the environment

1. **Double-click on `launcher.command`**
   A Terminal window opens with a numbered menu

2. **Type 1 and press Enter to Start**
   The launcher automatically starts Docker Desktop if it's not running, then launches all services

3. **Wait for the dashboard to open**
   The browser opens automatically at localhost:8082

! **Go to Portainer immediately!**
Open http://localhost:9000 within the first 5 minutes to create your account. See Ch.3 for details.

⏱️ **First time: wait 5 to 10 minutes**

Docker is downloading the images (PHP, MariaDB...). This is slow **only the first time**. Subsequent launches: less than 10 seconds.

💡 **Launcher menu**

The launcher offers 5 options: **1) Start**, **2) Stop**, **3) Restart**, **4) Reset Portainer**, **5) Quit**. Individual scripts (`start.command`, `stop.command`) are also available.

CHAPTER 3

3 Access the web interfaces

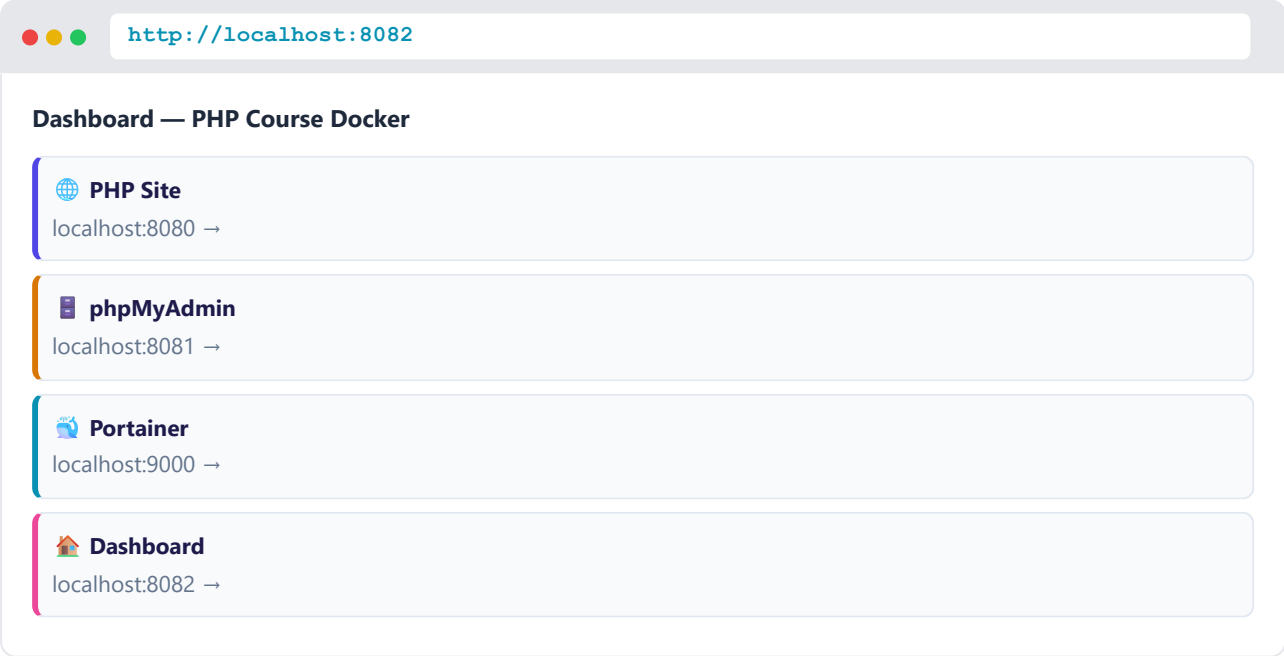
Everything happens in your browser

🗺️ Overview

SERVICE	ADDRESS	DESCRIPTION
🏠 Dashboard	`localhost:8082`	Home page with all links
🌐 PHP Site	`localhost:8080`	Your PHP files (htdocs/)
🗄️ phpMyAdmin	`localhost:8081`	Visual management of the DB
🐳 Portainer	`localhost:9000`	Docker container management

🏠 Dashboard — localhost:8082

Your daily **starting point**. It centralizes all links and your credentials.



🌐 PHP Site — localhost:8080

Displays the contents of your `htdocs/` folder. The test page confirms that PHP and the database are working.

http://localhost:8080

Hello PHP ! 👋

✅ MySQL connection successful! (MariaDB 10.6.x)

PHP 7.4.x — Linux 64-bit

🗄️ phpMyAdmin — localhost:8081

Graphical interface to manage your database. The connection is pre-configured (automatic).

🌐 Address	`http://localhost:8081`
👤 Username	`root`
🔑 Password	`root`
🗃️ Database	`app`

> ⏱️ **phpMyAdmin not responding?**
> The DB takes 10 to 20 seconds to start. Wait and refresh with **Cmd+R**.

🐳 Portainer — localhost:9000

Graphical interface to manage your Docker containers (start, stop, restart, view logs) — no command line needed.

> ⚠️ **Act within the first 5 minutes!**
> Portainer expires its initialization for security reasons. After startup, go **immediately** to http://localhost:9000 and create your account. If you see the "*timeout*" page → see the Troubleshooting chapter (p. 10).

Create your account (first access)

1. **Open http://localhost:9000**
   Portainer asks you to create an administrator account

2. **Enter a username and a password**
   Example: `admin` — Password: 12 characters minimum

3. **Click "Create user"**
   Then choose **"Get Started"** → **"local"**

What you can do in Portainer

http://localhost:9000 → Containers

Container	Status	Available actions
cours_web	running	▶ ■ ↺ 📋
cours_db	running	▶ ■ ↺ 📋
cours_phpmyadmin	running	▶ ■ ↺ 📋

▶ Start ■ Stop ↺ Restart 📋 View logs

CHAPTER 4

4 Working with PHP

Create, edit and test your PHP scripts

📁 The htdocs folder — your workspace

All your PHP files must be in the `htdocs/` folder of the project. It is the equivalent of the `htdocs` folder in XAMPP.

📁 Project structure

```
PHPenv/
├── 📁 htdocs/          ← YOUR PHP FILES HERE
│   ├── index.php       ← test page (already created)
│   └── hello.php       ← your next file
└── docker-compose.yml  ← do not modify
```

✍️ Create your first PHP file

1. **Open the `htdocs/` folder in Finder**
   Double-click on the folder from your Desktop

2. **Create a new file `hello.php`**
   In VS Code, TextEdit, or any text editor. The `.php` extension is required.

3. **Write your PHP code**
   Minimal example below

4. **Save and open in the browser**
   `http://localhost:8080/hello.php` — refresh with **Cmd+R**

```php
<?php
echo "Hello world!";
?>
```

🎉 **No need to restart!**

Every change to your PHP files is visible immediately after **Cmd+R** in the browser.

🔌 Database connection (PDO)

```
// Connect to the database
$pdo = new PDO(
    "mysql:host=db;dbname=app;charset=utf8mb4",
    "app",    // username
    "app"     // password
);

// Retrieve data
$stmt = $pdo->query("SELECT * FROM my_table");
foreach ($stmt as $row) {
    echo $row['name'] . "<br>";
}
```

💡 **The host is "db", not "localhost"**

In Docker, services communicate via their service name. The database is called `db`.

CHAPTER 5

5 Manage the database

Create tables and manipulate data with phpMyAdmin

🔑 Log in to phpMyAdmin

Go to **http://localhost:8081**. The connection is pre-configured, you are automatically logged in as root.

🌐 Address	`http://localhost:8081`
👤 Username	`root`
🔑 Password	`root`
🗄️ Database	`app`

🛠️ Create a table

1. **In the left column, click on "app"**
   This is your database

2. **"Structure" tab → "Create table"**
   Give it a name (e.g.: `users`) and choose the number of columns

3. **Define your columns, then "Save"**
   Always add an `id` column as INT AUTO_INCREMENT as the primary key

📊 Example "users" table

COLUMN	TYPE	LENGTH	OPTIONS
`id`	INT	11	PRIMARY KEY, AUTO_INCREMENT
`name`	VARCHAR	100	—
`email`	VARCHAR	150	—
`date_created`	DATETIME	—	DEFAULT CURRENT_TIMESTAMP

🔍 Execute an SQL query

"SQL" tab in phpMyAdmin:

```
-- Insert data
INSERT INTO users (name, email)
VALUES ('Alice Martin', 'alice@example.com');

-- Read all data
SELECT * FROM users;
```

CHAPTER 6

6 Stop and restart

Manage the environment without the command line

⏹ Stop the environment

Option A — Launcher (recommended)

Double-click on `launcher.command` → select **"Stop"**. You can also Restart or Reset Portainer from this menu.

Option B — Via Portainer

1. Open `localhost:9000`
2. **"Containers"** tab
3. Check all containers
4. Click **"Stop"**

💾 **Your data is preserved!**

Stopping the services does not delete your PHP files or your database data. Everything is preserved for the next launch.

↺ Restart a specific service

1. **Portainer → http://localhost:9000 → "Containers"**
2. **Find the container to restart**
   `cours_web` = Apache+PHP | `cours_db` = database | `cours_phpmyadmin`
3. **Click the ↺ (Restart) button**
   The service restarts in a few seconds

📋 View the logs

In Portainer → Containers → click on the **name** of a container → "Logs". Useful for diagnosing errors.

💡 **Daily routine**

Morning: Double-click `launcher.command` → type **1** (Start). Docker Desktop is launched automatically if needed.
Evening: `launcher.command` → type **2** (Stop)

CHAPTER 7

7 Mac Troubleshooting

Solutions to the most common issues

Common issues

❌ "Cannot be opened because it is from an unidentified developer"

Cause: macOS Gatekeeper blocks files downloaded from the internet.
Solution: Run the setup script in Terminal:

```
cd ~/Desktop/PHPenv && bash install.sh
```

Alternative: Right-click on the `.command` file → **"Open"** → click **"Open"** in the dialog. macOS remembers this choice.

❌ Portainer — "timeout" page (/timeout.html)

Cause: admin account not created within the first 5 minutes.
Solution: Use the launcher → option **4 (Reset Portainer)**, or run:

```
docker compose restart portainer
```

Then go **immediately** to http://localhost:9000 to create your account.

❌ Docker Desktop is not started

The launcher starts Docker Desktop **automatically**. If it fails, open **Docker Desktop** from the Launchpad → wait for the green whale → relaunch `launcher.command`

❌ localhost:8080 is not loading

Wait 30 seconds (the DB starts last) → **Cmd+R** to refresh.

❌ "Permission denied" on .command files

Run `install.sh` again, or open Terminal and type:

```
cd ~/Desktop/PHPenv && chmod +x *.command
```

❌ phpMyAdmin — connection error

The DB takes 10 to 20 seconds to start → wait and refresh (**Cmd+R**).

🔑 Credentials summary

🌐 PHP Site	`http://localhost:8080`
🗄️ phpMyAdmin	`http://localhost:8081`
🐳 Portainer	`http://localhost:9000`
🏠 Dashboard	`http://localhost:8082`
📁 DB Host (in PHP)	`db`
🗃️ Database	`app`
👤 DB Username	`app`
🔑 DB Password	`app`
👑 Root password	`root`

Mac Tutorial — Apache 2.4 + PHP 7.4 + MariaDB 10.6 + phpMyAdmin 5.2 + Portainer CE
macOS 14+ — Mac Intel & Apple Silicon (M1/M2/M3/M4/M5)